

IIN 3502 Distributed Network

Lab 01

Index No. – ICT/23/810

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1. Which devices in your topology represent the access layer, the distribution layer, and the core/edge layer, and what is the primary responsibility of each?

Access Layer: AccessSW1 and AccessSW2. Their main job is to connect the end devices (like the PCs) to the network.

Distribution Layer: DistSwitch. Its main job is to collect all the traffic from the access switches and do routing (Layer 3). It acts like a middle-man.

Core Layer: EdgeRouter. Its main job is to connect our whole campus network to the outside world (like the internet) and move traffic very fast.

2. Why was a Layer-3-capable switch chosen for the distribution layer instead of a standard Layer 2 switch?

I chose a layer 3 switch because it can do routing. It can move data between different networks (VLANs). A standard layer 2 switch only uses MAC addresses and cannot do routing. So we need a layer 3 switch at the distribution layer to handle the traffic between different parts of the network.

3. What would happen to the failure domain of this network if AccessSW1 failed? How could you redesign the topology to reduce this impact?

If AccessSW1 fails, PC1 and PC2 will lose their connection to the network. PC3, PC4, and the rest of the network will still work fine. To reduce this impact, we can redesign it by adding a second cable (redundant uplink) from AccessSW1 to DistSwitch, or by adding a backup switch. This way, if one link or switch fails, the PCs still have another path to the network.

4. If this campus grew to 10 access switches and 200 end devices, what changes would you make to the design (e.g., adding a core layer, redundant uplinks, PoE)?

- Add a dedicated core layer (separate from Distribution)
- Add redundant uplinks and EtherChannel for more bandwidth
- Use Power over Ethernet (PoE) switches if adding IP phones or wireless access points
- Implement dynamic routing protocols like OSPF or EIGRP

5. What criteria (port density, PoE support, uplink speed, routing capability, cost) would you use to select real hardware models for each layer in a production deployment?

- Port density (how many devices need to connect)
- PoE support (for phones/APs)
- Uplink speed (1G, 10G, 40G)
- Routing capability (layer 3 vs layer 2)

6. In Simulation mode, describe the path a frame from PC1 to PC4 took through the topology, and identify at which device(s), if any, Layer 3 processing would be required if these devices were on different VLANs.

PC1 > AccessSW1 > DistSwitch > AccessSW2 > PC4

If PC1 and PC4 were on different VLANs, the DistSwitch would do layer 3 processing. It would route the packet from one VLAN to the other. The Access switches would still only do Layer 2 switching.